Security of RSA Character-by-Character Encryption

27. Write a C program for Bob uses the RSA cryptosystem with a very large modulus n for which the factorization cannot be found in a reasonable amount of time. Suppose Alice sends a message to Bob by representing each alphabetic character as an integer between 0 and 25 (A S 0, c, Z S 25) and then encrypting each number separately using RSA with large e and large n. Is this method secure? If not,describe the most efficient attack against this encryption method.

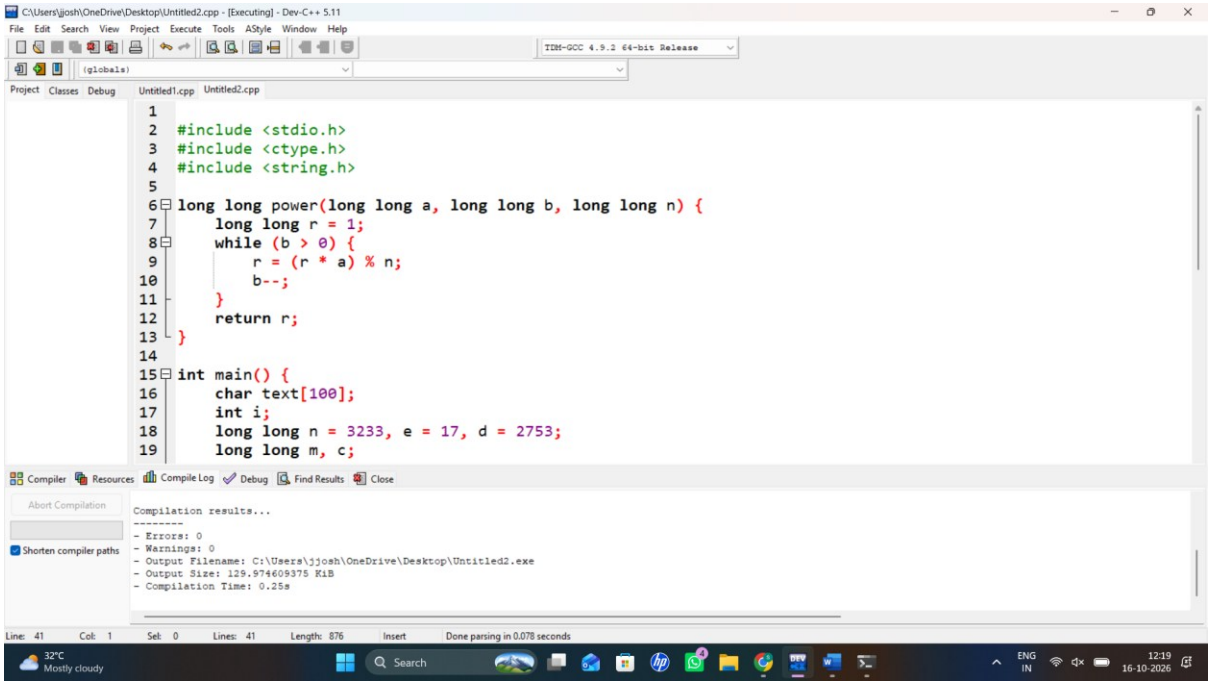
Aim:

To implement RSA encryption of alphabetic characters individually and demonstrate why this method is insecure.

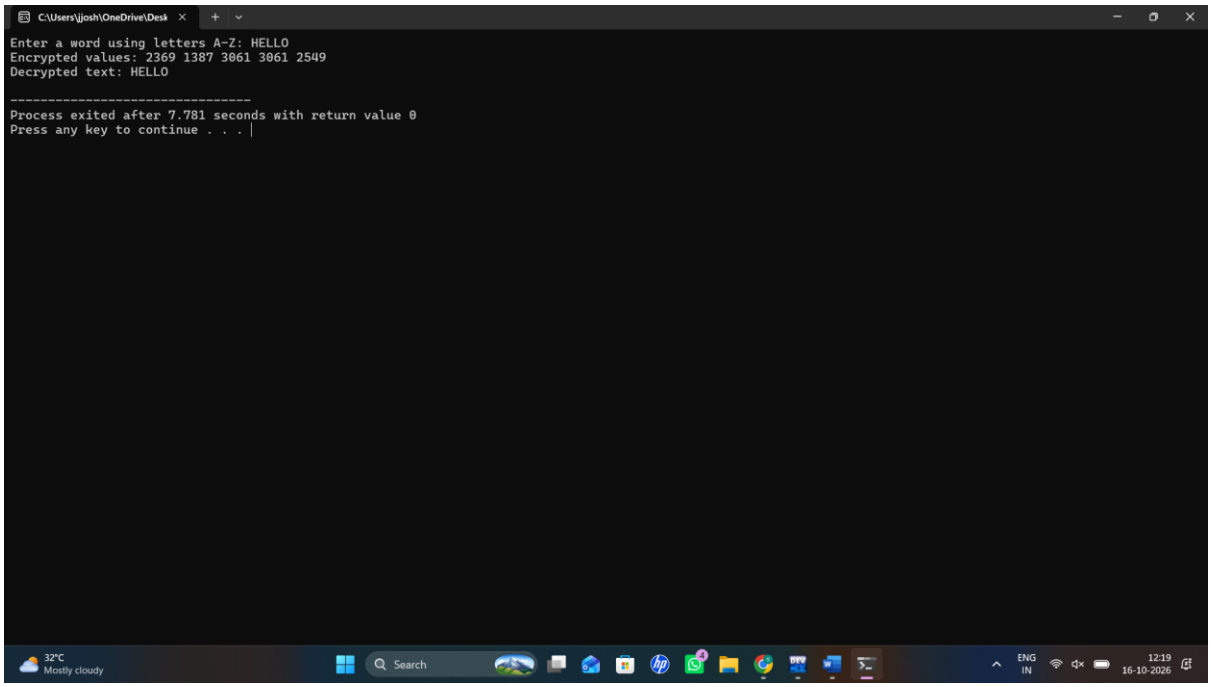
Algorithm:

1. Choose two prime numbers and calculate \(n\) and \(\phi(n)\).
2. Select public exponent \(e\) and private exponent \(d\).
3. Convert each character into an integer from 0 to 25.
4. Encrypt each integer using \(C=M^e\bmod n\).
5. Decrypt each ciphertext using \(M=C^d\bmod n\).
6. Observe that repeated characters produce repeated ciphertexts.

Program:



Input/Output:



Result:

The program demonstrates RSA character encryption and decryption. This method is not secure because RSA without randomized padding is deterministic. The most efficient attack is a frequency/dictionary attack: an attacker can compare ciphertext patterns, exploit repeated letters, and test the 26 possible character values against the public key. Real RSA should use secure randomized padding, such as RSA-OAEP, rather than encrypting individual characters.